Math 110 – Homework 12 Answers

Math 110 – Fall 2026

Homework 12

Positive operators, isometries, polar decomposition, and singular values

Suggested completion date: Friday, November 20

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Classify each of the following real symmetric matrices as positive definite, positive semidefinite but not positive definite, or not positive semidefinite:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Problem 2. Find the positive square root of

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Your answer should be an explicit 2×2 matrix.

Problem 3. Let $S : V \to W$ be linear between finite-dimensional inner-product spaces, and let $e_1, \ldots, e_n$ be an orthonormal basis of V. Prove that S is an isometry if and only if

$$Se_1, \ldots, Se_n$$

is an orthonormal list in W.

Problem 4. Find the polar decomposition

$$A = S\sqrt{A^*A}$$

of

$$A = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

acting on $\mathbb{R}^2$ with its standard inner product.

Problem 5. Find a singular value decomposition of

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Give the singular values and orthonormal bases e_1, e_2 and f_1, f_2 such that

$$Ae_j = s_j f_j, \qquad j = 1, 2.$$

Problem 6 (optional). Let T be an invertible operator on a finite-dimensional inner-product space. Prove that the singular values of T^{-1} are the reciprocals of the singular values of T.

Problem 7 (optional). Let T have singular values $s_1 \geq \cdots \geq s_n \geq 0$. Prove that

$$\|Tv\| \leq s_1 \|v\|$$

for every v, and show that equality occurs for some nonzero v.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes